# Beyond the edge of stability: periodic windows, quasi-periodicity and phase locking in the learning-rate plane of gradient descent

Authorship to be finalized by the human researcher

Private working draft, not for distribution

**AI-use disclosure.** Generative-AI tools helped implement, audit, execute, interpret, visualize, review, and draft this study. The author selected the research question, authorized the compute, has inspected the artifacts, and is responsible for the final text and claims. This is an independent, non-peer-reviewed preprint. Verify numbers against the released receipts before relying on them.
*Draft note: every clause above must be confirmed by the author before release.*

**Abstract**

Full-batch gradient descent with a large learning rate can oscillate, cycle or behave chaotically instead of settling. Trainability maps over learning-rate planes record a binary convergent-or-divergent label, which conflates these dynamical regimes. We resolve the dynamics behind such maps in two small tanh networks trained with separate learning rates for their two layers, and we organize the results around one thesis: beyond the edge of stability, the period-2 oscillation gives way to quasi-periodic training and phase-locked periodic windows.

(1) Confirmed in a prospectively designed and independently audited study: the learning-rate planes contain 61 numerically certified attracting periodic orbits of the full parameter-space update map, with periods from 4 to 58, whose domains are surrounded mainly by rerun-confirmed chaos. In six coarse maps, 87.1% to 99.97% of the cells that a published binary convergence score labels convergent were not classified as fixed points after 20,000 updates.

(2) Exploratory: on all 35 traced slices that start on the edge-of-stability period-2 orbit, that orbit loses stability as a complex pair of two-step multipliers crosses the unit circle, a Neimark–Sacker candidate. On one slice the crossing angle is 0.2180 turns per two steps and the first quasi-periodic cell rotates at 0.2180. Of 56 eligible windows, 35 pass a Birkhoff ordering test of their projected orbits, against at most 3.58 expected by chance. Only 2 of 61 windows meet draft criteria for classical shrimp anatomy.

Takeaway: the region that a binary map calls trainable holds certified periodic windows inside chaos, and exploratory evidence ties them to quasi-periodic training born where the period-2 orbit acquires complex multipliers. The claim we test next is local: a period-2 branch undergoes an attracting Neimark–Sacker bifurcation and a rule-selected window is connected to its resonance by continuation. Nothing here concerns stochastic or large-scale training or establishes a training benefit.

**Study status.** Results labeled CONFIRMED (PROSPECTIVELY DESIGNED, INDEPENDENTLY AUDITED) come from the map study of Section 4, whose design, thresholds and analysis were pushed before outcomes and repeated by an independent audit. Results labeled EXPLORATORY (Section 5) were found while exploring and have not been tested on held-out networks. Provenance for every figure and number is described in Section 9.

# 1 Introduction

Training with a learning rate beyond the classical stability limit is common practice. At the edge of stability, the sharpness of the loss hovers near $2/\eta$ while the loss still decreases over long horizons [6]; self-stabilization explains how sharpness is held there [9], progressive sharpening how it gets there [1], and the catapult phase what happens to the loss at somewhat larger rates [21]. Simplified models show period doubling and chaos as the step size grows [5, 13, 18]. Over two-dimensional hyperparameter planes, the boundary between trainable and divergent settings is fractal [22, 23, 28, 31]. These maps record whether the loss falls and stays bounded, not what the parameters do.

We ask what the dynamics inside such maps are. We treat full-batch gradient descent with separate learning rates for the hidden and output layers as a two-parameter family of maps on parameter space. For every cell of a learning-rate grid we screen the parameter state at the end of a long run (fixed point, periodic orbit with its period, chaos, drift, divergence), certify selected periodic orbits with Newton's method on the exact Jacobian, and follow orbits and their stability boundaries through the plane.

**Thesis and status.** We organize the results around one thesis: beyond the edge of stability, the period-2 oscillation gives way to quasi-periodic training on an invariant circle, rational rotation numbers lock into periodic windows, and the windows of the learning-rate plane are largely these locked orbits. Each result below says exactly what passed. The certified windows and the comparison with the binary score are confirmed. The complex-pair crossings, rotation numbers, ordering tests and staircases are exploratory; we call the crossings Neimark–Sacker candidates until transversality, nondegeneracy and an attracting invariant circle are documented. The claim we will test on held-out networks is local and existential: in specified finite tanh networks under deterministic full-batch gradient descent in float64, a period-2 branch undergoes an attracting Neimark–Sacker bifurcation and a rule-selected window is connected to the corresponding resonance by continuation.

**Prior work.** Periodic windows, period doubling, positive Lyapunov exponents and broadband power spectra in learning dynamics, with the learning rate as bifurcation parameter, date back at least to van der Maas et al. [32], who studied a three-unit autoassociator. Their delay-embedded chaotic attractors are thin folded curves, effectively one-dimensional chaos of the period-doubling kind, unlike the invariant circles of the two-step map studied here. Period-colored maps over optimizer hyperparameters have been drawn from a two-dimensional projection of the error, without full-state verification [30]. Hofmann [13] studies finite-step gradient descent as a dynamical system in solvable models: the edge of stability as the first (flip) bifurcation of the training map, the period-doubling route with windows, and, in a two-mode depth-two linear network, a "rotational edge" beyond which the unstable multipliers of the aligned 2-cycle are complex and the system saturates in a persistently misaligned oscillatory state. Our analysis starts where that result stops: we ask whether such a crossing produces quasi-periodic training whose rotation is set by the crossing multiplier, and whether periodic windows are its resonances. Riddled basins and learning-rate-dependent reproducibility are documented by Ly and Gong [24]. Invariant curves and phase-locking Arnold tongues appear in a multiplicative-weights learning rule for population games [25], and chaos in learning a two-person game goes back to Sato et al. [27]; both concern games rather than descent on one loss. Heavy-ball momentum has cycles that rotate by $1/K$ [12].

**This paper's contribution.**

- Confirmed: a period-resolved description of the learning-rate plane of full-batch gradient descent in two small networks, with 61 certified attracting windows embedded in chaos, repeated by an independent audit.
- Confirmed: a cell-by-cell comparison with the published binary convergence score, which conflates fixed points, periodic orbits, chaos and drift.
- Exploratory: complex-pair crossings of the period-2 orbit along a traced curve, rotation numbers of the quasi-periodic training that follows, a Birkhoff ordering test of the windows against a null, and rotation-number staircases.
- Code, receipts and a verification script that re-derive every figure and number.

**Not claimed.**

- No Neimark–Sacker bifurcation in the strict sense: transversality, nondegeneracy and an attracting invariant circle are not yet documented.
- No route to chaos: no prespecified connected path through sustained chaos has been tested.
- No proof that the windows are Arnold tongues: the Birkhoff test is a diagnostic of projected orbits, and resonance ancestry is pending.
- No claim about stochastic, minibatch or large-scale training, other architectures or real data, and no training benefit.

## 2 Gradient descent as a family of maps

Let $L(\theta)$ be the full-batch training loss of a network with parameters $\theta \in \mathbb{R}^n$, split into hidden-layer and output-layer blocks. Gradient descent with per-layer learning rates is the map

$$\theta_{t+1} = F(\theta_t) = \theta_t - P\,\nabla L(\theta_t), \qquad P = \mathrm{diag}(\eta_h \mathbf{1}_{\text{hidden}}, \eta_o \mathbf{1}_{\text{output}}), \tag{1}$$

a family of maps indexed by the two rates $(\eta_h, \eta_o)$. Its Jacobian is $DF(\theta) = I - PH(\theta)$ with $H$ the Hessian of $L$. An orbit of period $p$ is stable when all eigenvalues (multipliers) of $DF(\theta_{p-1}) \cdots DF(\theta_0)$ lie inside the unit circle. Generically it loses stability when a multiplier crosses the circle: at $+1$ (fold), at $-1$ (flip, which doubles the period) or as a complex pair $e^{\pm 2\pi i\omega}$ (Neimark–Sacker, which under transversality and nondegeneracy conditions creates an invariant closed curve) [20].

**Why the candidates start at period 2.** $DF(\theta)$ is similar to the symmetric matrix $I - P^{1/2} H P^{1/2}$, so at a fixed point all multipliers are real and a fixed point loses stability by fold or flip only. The edge-of-stability oscillation is the period-2 orbit born at that flip. For a period-2 orbit $\{\theta_a, \theta_b\}$, the two-step Jacobian is similar to $S_b S_a$ with $S_j = I - P^{1/2} H(\theta_j) P^{1/2}$. A product of two symmetric matrices has real eigenvalues when one factor is positive definite, but at the edge of stability each $S_j$ has a negative eigenvalue in the oscillating direction, so complex multipliers become possible. This is an elementary observation; complex unstable multipliers of an edge-of-stability 2-cycle were derived for a two-mode linear network by Hofmann [13].

**Circle maps and phase locking.** On an invariant circle the dynamics has a rotation number $\rho$, the mean fraction of a turn per iterate. When $\rho = m/q$ is rational the motion can lock onto a periodic orbit; the locked parameter sets are Arnold tongues, rotation numbers along a parameter path form a devil's staircase, and the widest plateau between two locked rationals is typically their Farey mediant [2, 14, 15]. Breakdown of the circle leads to chaos [3, 8, 26]. Because the circle in question belongs to the two-step map, a locked orbit at rotation $m/q$ has period $2q$ for $F$; this relation is a consequence of the setting, not evidence.

**Shrimps.** In two-parameter families of dissipative maps, periodic windows inside chaos often take a characteristic "shrimp" shape bounded by fold and flip curves [10, 11]. We test for this anatomy explicitly rather than assume it.

# 3 Methods

| Network | Parameters | Training points | Architecture | Weight decay | Seeds | Maps |
|---|---|---|---|---|---|---|
| S1 | 72 | 72 standard-normal | 8 inputs, 8 tanh units, no biases, mean-field | 0 | 28001, 28002 | 2 coarse |
| S1R | 72 | 72 standard-normal | 8 inputs, 8 tanh units, no biases, mean-field | $10^{-3}$ | 28001, 28002 | 2 coarse, 2 zoom |
| S2 | 13 | 24 from a tanh teacher | 2 inputs, 3 tanh units, biases | $10^{-3}$ | 28011, 28012 | 2 coarse, 2 zoom |

**Table 1: Design of the map study.** Every map has $256 \times 256$ logarithmically spaced pairs of hidden- and output-layer rates; coarse maps span 3 decades in each rate and zoom maps 0.36 decades. Every cell runs 20,000 full-batch updates in float64 from its seed's initial parameters. S1 is the published configuration of the fractal-trainability study and is reported descriptively.

**Networks and data.** Table 1 lists the design. S1 reproduces the published mean-field configuration of the fractal-trainability study [28]; S1R adds weight decay; S2 is a 13-parameter tanh regressor fitted to points from a random tanh teacher. We refer to a network and seed together, for example S2-28011. Training is deterministic full-batch gradient descent in float64.

**Three different quantities.** The published binary score measures loss reduction: a run counts as convergent when its loss, normalized by the initial loss and averaged over the last 20 of the first 1,000 updates, is below one [28]. Our screening labels classify the parameter state at the end of a finite run. Certification validates individual periodic orbits. The three need not agree, and we compare them cell by cell.

**Screening.** Every cell runs 20,000 updates from the seed's initial parameters. The last 384 updates are screened for the smallest full-state recurrence period up to 64: a period $p$ requires the state to recur with componentwise tolerance $10^{-8} + 10^{-6}\,|\theta|$ over the whole tail, which spans at least 6 repeats. The largest finite-time Lyapunov exponent (FTLE) is averaged over the last 4,000 updates after 1,000 updates of tangent alignment. Labels are fixed point, periodic, chaotic (FTLE above 0.005 per update), drift (parameter norm still growing by more than 1% over the FTLE window), unresolved, or divergent (non-finite or larger than $10^8$).

**Numerically certified.** An orbit of period $p$ is numerically certified when damped Newton iteration on $F^p(\theta) - \theta = 0$, using the exact chain-rule Jacobian of $F^p$, reaches an undamped relative correction below $10^{-10}$ within 60 iterations; the converged point closes to within $10^{-9} + 10^{-8}\,|\theta|$ in every component; it lies within $10^{-3}$ (relative) of the screened candidate; every proper-divisor iterate is separated from it by at least $10^{-6}$ (relative), so the period is primitive; and it is attracting when the spectral radius of the monodromy matrix is below $1 - 10^{-7}$. This is a floating-point verification, not a computer-assisted proof.

**Embedded windows and audit.** A connected component of equal-period cells qualifies as an embedded window when its period is at least 3, it has at least 4 cells, no cell of its exterior ring is a fixed point or has a proper-divisor period, at least 50% of the ring is rerun-confirmed chaos (reruns of 30,000 updates with a 10,000-update FTLE and a different tangent vector, up to 48 ring cells per component), and its representative orbit is certified and attracting. The audit was a separate program that imports neither the batched engine, the orbit certifier nor the selection helpers: it rewrote the one-cell map, the recurrence screen, the zoom rule and the component and ring rules, rechecked every certificate with finite-difference Jacobians (all 154 of 154 pass closure, primitivity, stability and seed checks; spectral radii agree within $1.7 \times 10^{-7}$), re-simulated the ring cells of every qualifying component, recomputed the gate on its own evidence, and found 0 mismatches among the 276 receipt files.

**Continuation and boundary type.** From the center of a window, its certified orbit is continued along 64 rays in the rate plane until it loses stability, located by bisection, and the boundary is typed by the crossing multiplier: fold ($+1$), flip ($-1$) or complex pair. We do not check transversality or nondegeneracy, so all three types are candidates; we write "Neimark–Sacker candidate" for the complex-pair case because the thesis rests on it. The same procedure traces where the period-2 orbit loses stability along slices of fixed output rate.

**Rotation numbers and the Birkhoff test.** For a non-periodic cell we take the two-step samples $z_k = \theta_{2k}$, project them on their first two principal axes, and define the rotation number as the mean unwrapped angle increment in turns per two steps. For a certified orbit of period $p$ we take its two-step points ($q = p/2$ points for even $p$, $q = p$ for odd $p$), sort them by angle in their principal plane, and say the orbit passes when every step advances the angular rank by the same shift $m$ with $\gcd(m, q) = 1$, the ordering of a Birkhoff orbit of rotation $m/q$. The number of points $q$, and therefore the relation $p = 2q$, is fixed by the period; what the test measures is the ordering and the shift. Orbits with 2 or 3 points pass by construction. The null is the pass probability of the same test for uniformly random angles; for a group of orbits we report the expected number of chance passes. The refined test also orders points along the shortest closed tour through them and tests the points of $F^{2^{k+1}}$ ($k = 1, 2, 3$), which can detect orbits born by period doubling inside a locked window; it counts only tests on at least 5 points, and its null for each orbit is the sum of the per-test pass probabilities (a union bound), so the summed null is an upper bound on the expected number of chance passes.

## 4 Periodic windows inside chaotic training

Status: CONFIRMED (PROSPECTIVELY DESIGNED, INDEPENDENTLY AUDITED).

The learning-rate planes of S1R and S2 contain 61 embedded windows that qualify on both the producer's and the auditor's evidence (62 and 63 respectively; the disagreements all sit at the 50%

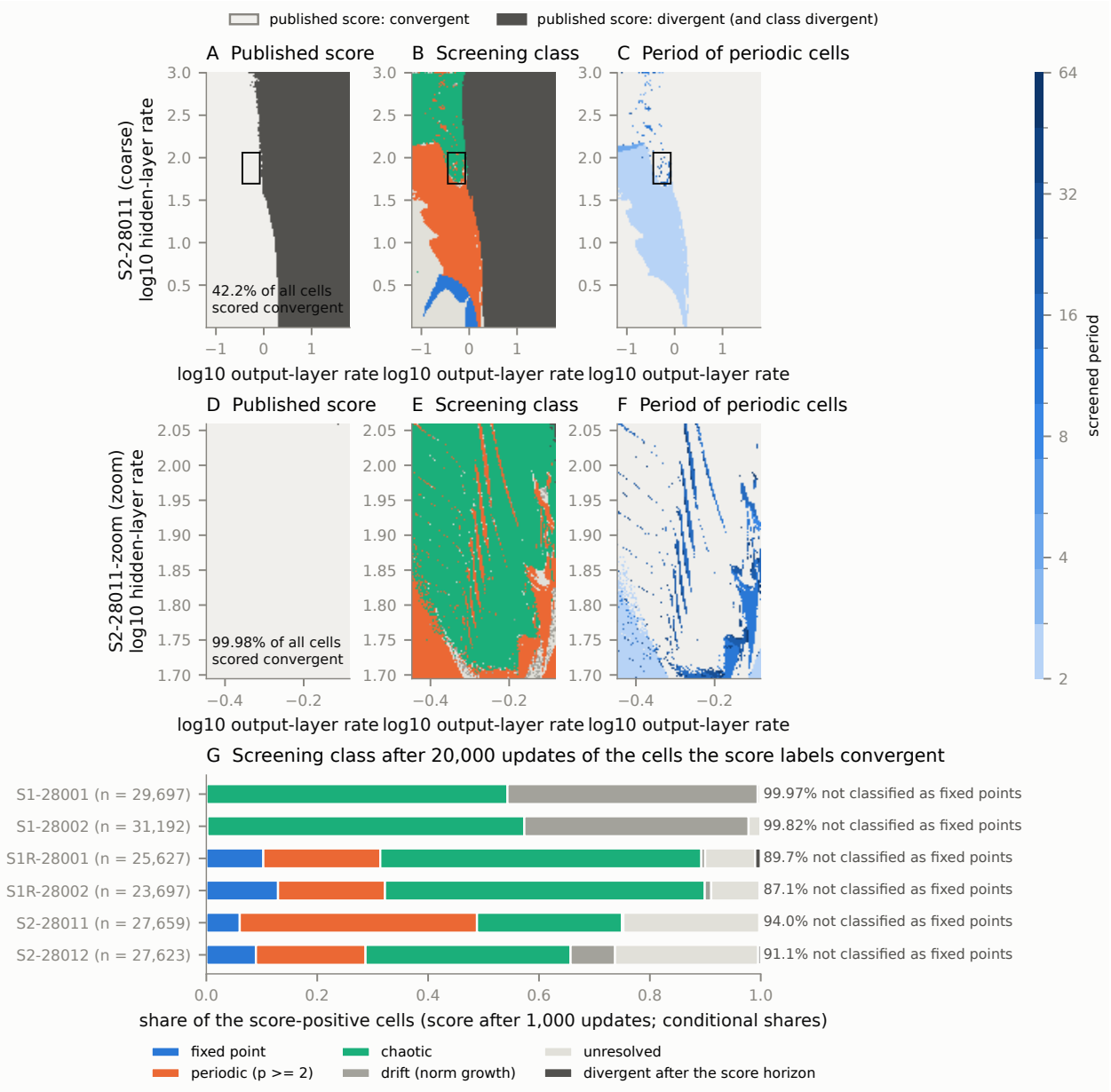



**Figure 1: The binary score conflates dynamical regimes: the cells it labels convergent carry fixed-point, periodic, chaotic and drift labels after 20,000 updates.** Panels A and D show the published score after 1,000 updates; the printed share is the fraction of all cells in that map scored convergent. Panels B, C, E and F show the screening class and, for periodic cells, the period of the same cells after 20,000 updates (S2-28011, coarse and zoomed; the rectangle marks the zoom). Panel G is conditional: among the cells the score labels convergent, the share in each screening class, for all six coarse maps. Screening labels are finite-time labels; only certified orbits are verified. *Generated by* `paper/gd-route/figures/f1_binary_vs_period.py`; *receipt* `figures/f1_binary_vs_period.provenance.json`.

ring threshold). All 154 attempted orbit certifications verified. Confirmed periods range from 4 to 58 and include the odd periods 5, 11, 13, 17 and 29. The exterior rings were checked with 2,801 reruns, and 157 of 160 randomly chosen chaotic cells (16 per map) stayed chaotic on a rerun with a different tangent vector.

Figure 1 compares these dynamics with the published binary score. In the six coarse maps, 87.1% to 99.97% of the cells the score labels convergent were not classified as fixed points after 20,000 updates; the score's success means the loss fell, not that the parameters settled. Task success and non-convergent learning dynamics can coexist: van der Maas et al. [32] reported that their network performed its task across the whole learning-rate range they studied, including where learning was chaotic. The published configuration without weight decay (S1) has 0 periodic cells in both seeds: its runs diverge, stay chaotic, or drift toward large weights while the loss slowly falls. Adding weight decay makes the windows appear.

# 5 Exploratory evidence for the thesis

Status: EXPLORATORY. The analyses in this section were chosen while exploring and used data that shaped the thesis.

## 5.1 The period-2 orbit acquires complex multipliers: Neimark–Sacker candidates

We continued the period-2 orbit of S2-28011 toward larger hidden-layer rates along 40 slices of fixed output rate. On all 35 slices that started on it, the orbit loses stability as a complex pair of two-step multipliers crosses the unit circle (Figure 2). The crossing curve is U-shaped, with its lowest point at $\log_{10}\eta_h = 1.650$, and the crossing angle $\omega$ ranges from 0.1670 to 0.2401 turns per two steps, passing the resonances 2/11, 1/5, 2/9 and 3/13.

On the slice through the period-18 window, the curve interpolated to that slice places the crossing at $\log_{10}\eta_h = 1.738$ with $\omega = 0.2180$; in an independent staircase computation the first quasi-periodic cell lies at 1.738 and rotates at 0.2180. Agreement of the rotation number with the crossing angle is what a Neimark–Sacker bifurcation predicts, but it does not establish one: the transversality of the crossing, the nondegeneracy (first Lyapunov) coefficient and an attracting invariant circle with converged residuals remain to be documented.

**Placeholder.** Rotation-number map of the S2-28011 plane overlaid with the crossing curve, testing whether the rotation just beyond the curve equals the crossing angle everywhere along it (computation in progress).

**Placeholder.** Resonance ancestry: continuation of a rule-selected window (the 1/5 window) through both of its fold edges back to the 1/5 resonance on the crossing curve, with the onset, circle and detuning acceptance rules frozen before any confirmation (in progress).

## 5.2 States along one slice

Figure 3 shows what training looks like along that slice. Beyond the crossing, the two-step samples lie on a closed curve (rotation 0.2159, FTLE 0.0001); inside the period-18 window they are nine points (rotation 0.2222, FTLE -0.0072); between the 2/9 and 1/4 plateaus the curve thickens and breaks (FTLE 0.0504).

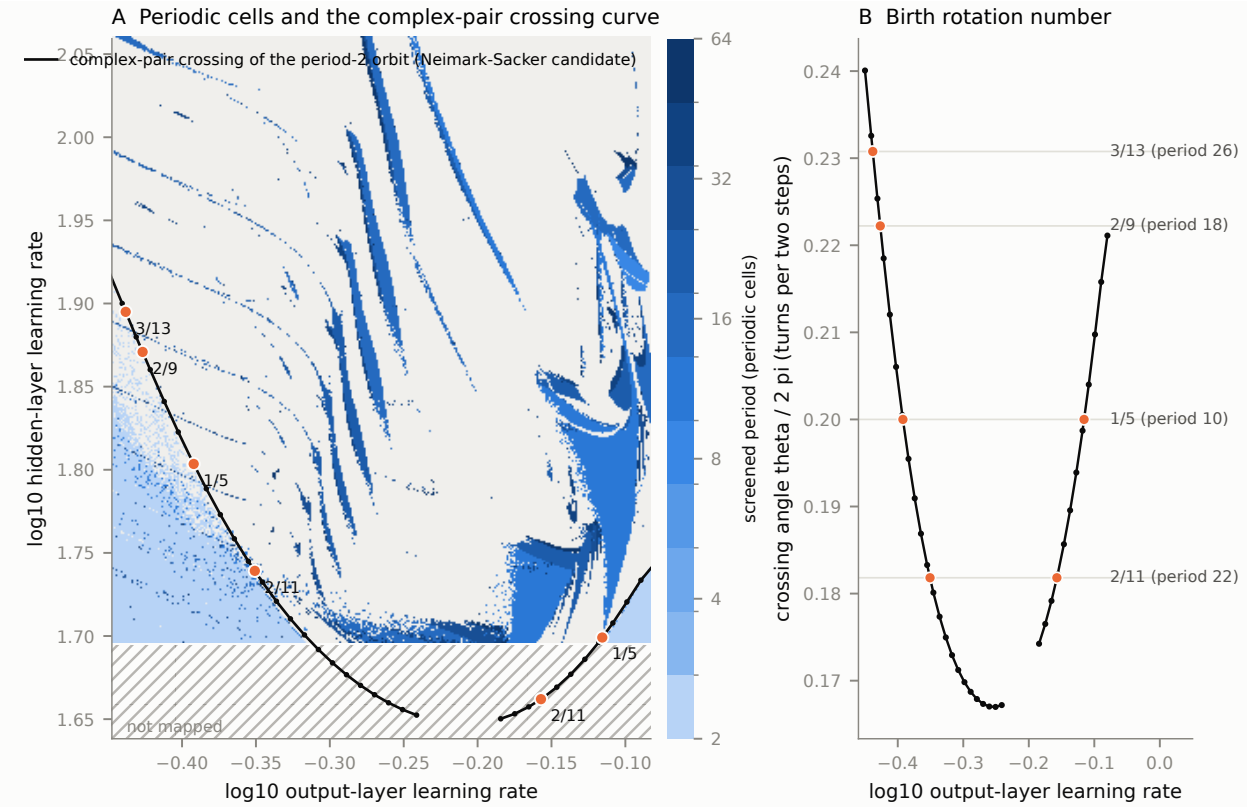



**Figure 2: Along the whole traced curve, the edge-of-stability period-2 orbit loses stability as a complex pair of two-step multipliers crosses the unit circle (Neimark–Sacker candidates).** A: periodic cells of a 1,024 × 1,024 map (shade: period), the traced crossing curve (black) and the resonance points where the crossing angle equals a rational $m/q$ with $q \leq 13$ (orange); tongues would be rooted there if the crossings are nondegenerate Neimark–Sacker bifurcations. B: crossing angle along the curve. On 5 of 40 slices the starting point was not on the period-2 orbit; those slices break the curve. Resonance points are interpolated between traced slices, and tongues near their roots are narrower than one map cell. *Generated by* `paper/gd-route/figures/f2_ns_tongue_roots.py`; *receipt* `figures/f2_ns_tongue_roots.provenance.json`.

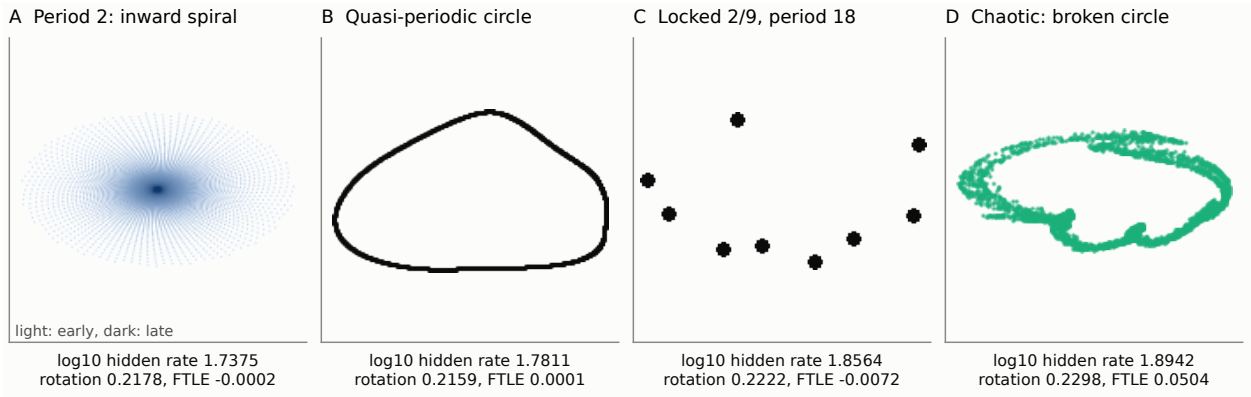



**Figure 3: Four states along one slice, in order of increasing hidden-layer rate: an inward spiral to the 2-cycle, a closed invariant curve, nine locked points and a broken curve.** Each panel shows every second parameter vector of one run projected on its own two principal axes. The rates are chosen by a fixed rule from the staircase receipt (the period-2 cell closest to the onset, the median quasi-periodic cell, the middle of the longest 2/9 plateau, the median chaotic cell between the 2/9 and 1/4 plateaus), not by eye. The states are not a monotone sequence: beyond the onset the rotation number first falls, and beyond panel D training locks again at 1/4 before chaos returns (Figure 4). Projections onto two axes can hide structure in the remaining dimensions. *Generated by* `paper/gd-route/figures/f6_route_portraits.py`; *receipt* `figures/f6_route_portraits.provenance.json`.

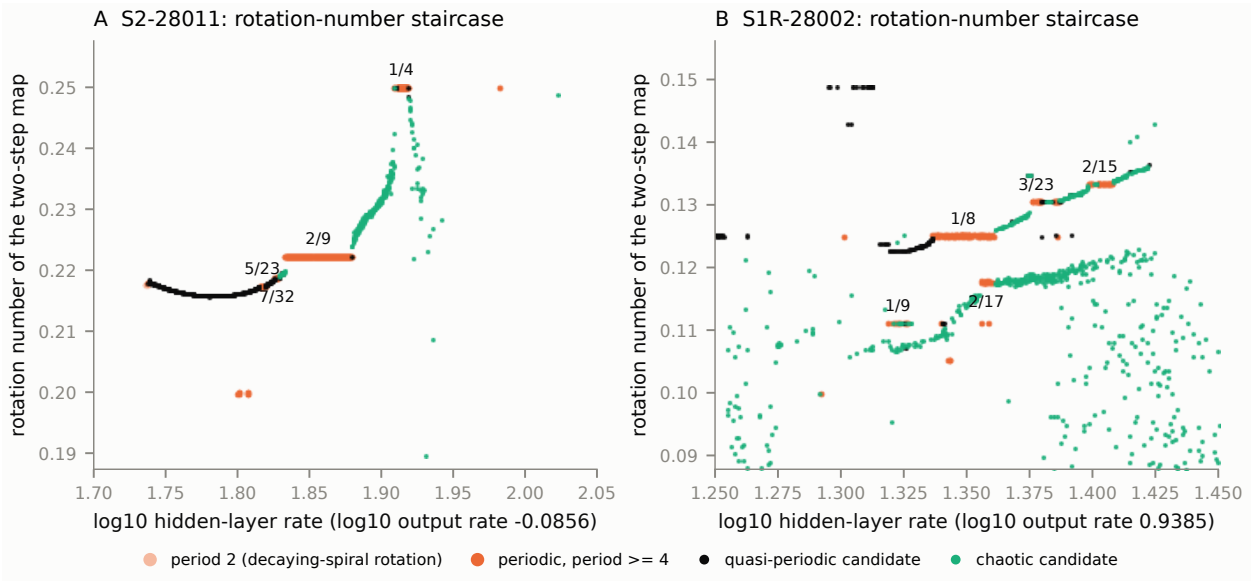



**Figure 4: Along slices, quasi-periodic cells trace continuous rotation numbers and periodic cells sit on rational plateaus.** Plateaus are detected automatically (locked cells with period $2q$ or a doubling of it, rotation within $6 \times 10^{-4}$ of $m/q$) and labeled when they hold at least 8 cells. In S2 the rotation number first falls after the onset, then rises; the cells between the 2/9 and 1/4 plateaus are almost all chaotic candidates. In S1R two rotation branches coexist over the same rates, a sign of coexisting attractors. Chaotic cells outside the plotted range are not shown. *Generated by* `paper/gd-route/figures/f3_staircases.py`; *receipt* `figures/f3_staircases.provenance.json`.

### 5.3 Rotation numbers form staircases

Along the slice through the period-18 window of S2-28011 (Figure 4), the rotation number falls from 0.2180 at the onset to 0.2158 at $\log_{10}\eta_h = 1.780$, then rises and locks at 2/9 for $\log_{10}\eta_h$ from 1.833 to 1.879. Of the 168 cells between the 2/9 and 1/4 plateaus, 167 are chaotic candidates; training then locks at 1/4 from 1.909 to 1.919 before chaos returns. The labeled plateaus are 5/23, 7/32, 2/9 and 1/4 in S2 and 1/9, 2/17, 1/8, 3/23 and 2/15 in S1R. Consecutive plateaus are Farey neighbors ($|m_1q_2 - m_2q_1| = 1$) in 3 of 3 pairs in S2, against 49% per pair for randomly drawn fractions in the same range, and in 4 of 4 pairs in S1R, where the null is 100% because every fraction with a small enough denominator in that range appears. The Farey pattern is weak evidence in S2 and uninformative in S1R.

### 5.4 A Birkhoff ordering test of the windows

Across the 61 windows, the 3,904 rays end at 1,316 folds, 1,012 flips and 1,395 complex-pair crossings (181 rays reached the end of their range or stayed unresolved; Figure 5A). Fold edges are what the edges of a locked window on a circle look like, while flips and secondary complex-pair crossings are expected where a locked orbit loses stability off the circle.

The Birkhoff test results depend on which orbits are eligible, so we give each denominator. Of the 129 certified orbits with period at least 3, 115 have at least 5 two-step points; 65 of these pass the plane test, against 1.57 expected by chance. All 61 confirmed windows are eligible for the plane test and 31 pass, against 4.92 expected; this count includes 4 windows with at most 3 points, which pass by construction. For the refined test, 56 confirmed windows have at least one test on 5 or more points (the 5 excluded windows have periods 4, 4, 6, 6 and 8); 35 of them pass, against a union bound of 3.58 chance passes. 28 pass at the two-step level, with measured shifts $m = 1$ for 19, $m = 2$ for 8 and $m = 3$ for 1; 7 pass first at a period-doubled level, for example period-20 orbits ordered under four steps with rotation 2/5, twice 1/5. Of the 5 odd-period windows, 0 pass (null 0.17): they did not pass this diagnostic, and their organization is open.

### 5.5 Classical shrimp anatomy is the exception

Draft criteria for classical shrimp anatomy require fold and flip boundaries each on at least 8 of 64 rays, at least half of the flip rays opening onto period $2p$ in a local map, and agreement between ray and map edges on at least three quarters of rays. 15 windows meet the boundary-count part and 2 meet all of it. The clearest, the period-18 window of S2-28011 (Figure 6), has 46 fold, 15 flip and 3 complex-pair rays; ray and map edges agree within two cells on 61 of 64 rays, and 10 of its flip rays open onto period 36. Complex-pair boundaries, absent from classical shrimps, are common: 41 of 61 windows have at least 8 such rays.

**Placeholder.** Minimal model bridge: the two-mode depth-two linear network of Hofmann [13] beyond its rotational edge, testing whether the persistently misaligned oscillatory state is quasi-periodic, with a rotation number continuous from the crossing angle, and whether it has locking windows (in progress).

**Placeholder.** Prespecified route-to-chaos path: a connected path, fixed in advance, from the period-2 region through the quasi-periodic band and locked windows into sustained chaos with converged diagnostics (not yet run).

**Placeholder.** Bounded transfer to real data: a small network on a CIFAR subset, developed on a first subset with everything frozen before a second subset is opened (not yet run).

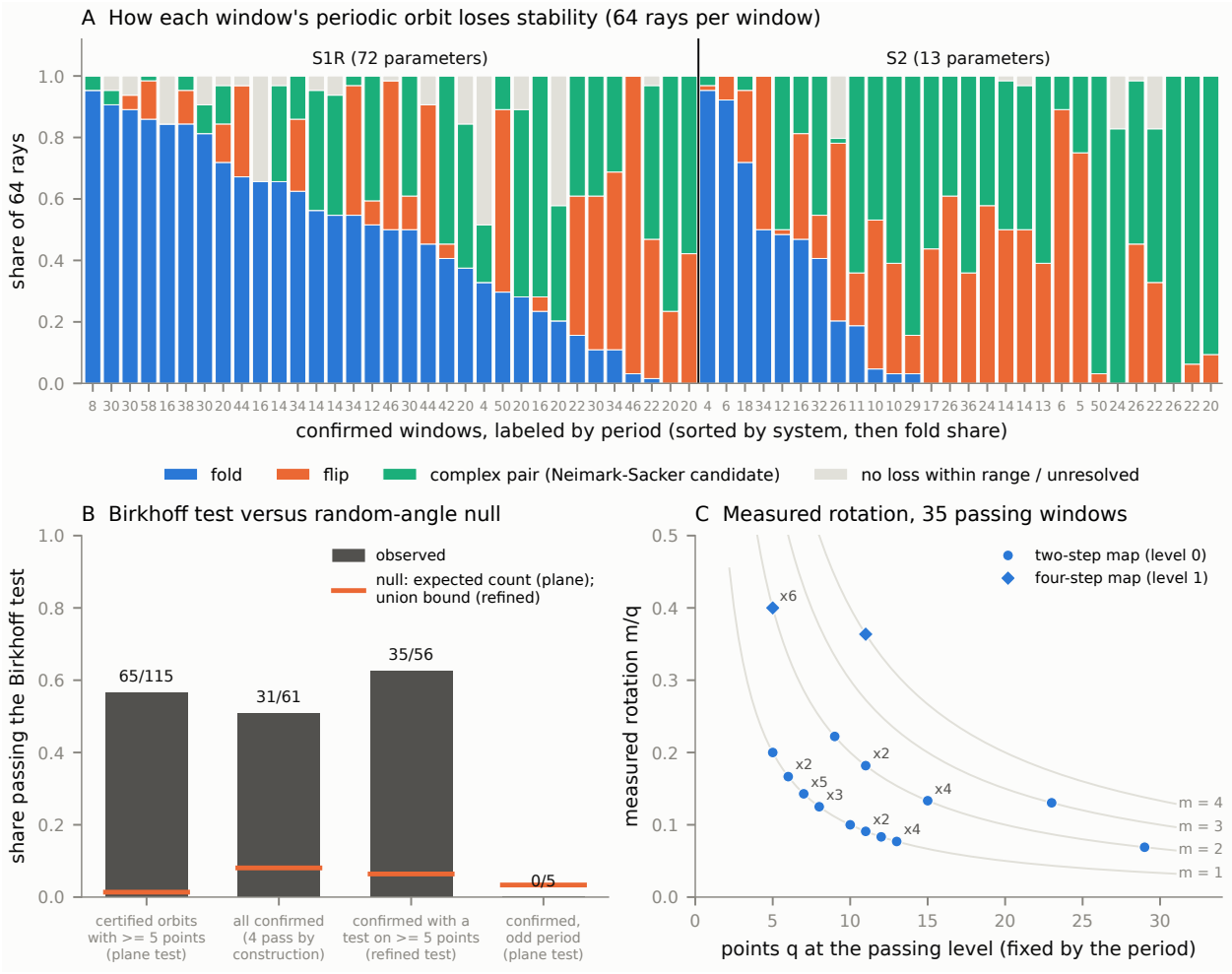



**Figure 5: Across all 61 windows, stability boundaries mix fold, flip and complex-pair crossings, and most eligible windows pass a Birkhoff ordering test far above its null.** A: boundary type along 64 rays per window. B: share of orbits passing the test (gray) against the null (orange): the expected share of chance passes for the plane test, and the union bound for the refined test; eligibility is printed under each bar. C: rotation $m/q$ measured by the passing test at its first passing level; $q$ is fixed by the period and level by construction, so only the shift $m$ and the ordering are measured. *Generated by* `paper/gd-route/figures/f5_window_census.py`; *receipt* `figures/f5_window_census.provenance.json`.

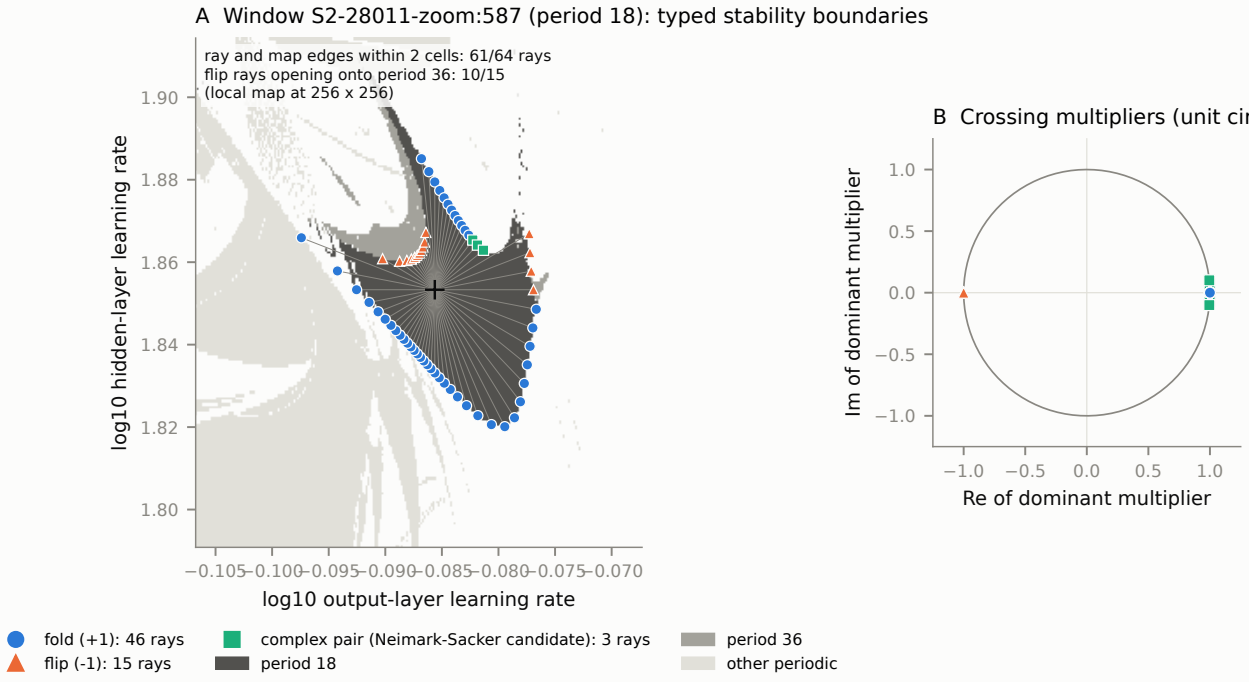



**Figure 6: The period-18 window of S2-28011 has a fold-bounded body with flip edges opening onto period-36 bands.** A: local map from the shared initial parameters with the window's orbit continued along 64 rays; markers show where and how it loses stability. B: crossing multipliers on the unit circle. This window was selected as the clearest example and is one of only 2 meeting the draft shrimp criteria; it is not representative (Figure 5). *Generated by* `paper/gd-route/figures/f4_window_anatomy.py`; *receipt* `figures/f4_window_anatomy.provenance.json`.

**Placeholder.** Consequences for training: the exploratory consequence pilot (reproducibility, stopping-time sensitivity and loss against regime, with periodic and chaotic controls, and incremental value beyond the Lyapunov exponent and sharpness), to be added with numbers generated from its receipts.

# 6 Discussion

Everything in this section is interpretation. The confirmed results in Section 4 stand on their own, and the hypotheses here need their own prespecified tests.

The picture that fits the exploratory evidence is the quasi-periodic organization familiar from circle maps and forced oscillators [3, 8, 15]: the edge-of-stability oscillation is a period-2 orbit; past a complex-pair crossing of the two-step map, training moves on an invariant circle with a rotation number set by the crossing angle; rational rotations lock into periodic windows; and the circle breaks, with period doubling inside locked windows along the way. Section 2 explains why the crossing needs at least period 2 in gradient descent and why per-layer rates are not required for it in principle; the rotational edge of Hofmann [13] is the same kind of crossing in a solvable model. Under this reading, most windows of Section 4 are locked orbits, and classical shrimps are a special case.

**Reading the rotation from the loss.** In exploratory analyses of the consequence pilot, the dominant peak of the even-step training-loss spectrum tracked the two-step rotation number, and an autoregressive fit to noisy loss recovered the period-2 orbit's complex multiplier before the crossing. We treat this as local system identification, not early warning, until sampling, harmonic, abstention, autoregressive-baseline and specificity rules are fixed in advance. The earliest spectral diagnostic of learning dynamics we know of is that of van der Maas et al. [32], who used broadband spectra to recognize chaotic learning; here a spectral peak would carry the rotation number.

**Open questions.** The odd-period windows did not pass the Birkhoff diagnostic; whether they are organized as in periodicity hubs [17] is open. Windows also coexist with other attractors, so maps started from one initial point show which attractor that point reaches rather than the full existence region of a window. And the evidence comes from two small networks.

**The test that would settle the local claim.** An existence-and-replication test would fix in advance a single primary endpoint (a bounded set of mechanism criteria: a simple transverse complex-pair crossing with all other multipliers inside the circle, nondegeneracy, an attracting invariant circle with converged residuals, rotation compared with the crossing angle at prespecified detunings, and ancestry of a rule-selected window through both folds), and apply it to two prespecified independent losses across two architectures, one of them unused so far, with no substitution.

# 7 Limitations

The networks have 13 and 72 parameters and are trained on synthetic data with deterministic full-batch updates in float64. Screening labels are finite-time; chaotic labels rest on single-tangent exponents and reruns. Certification is a floating-point verification. Rotation numbers and the Birkhoff test use two-dimensional projections of curves that live in 13 or 72 dimensions. The mechanism analyses were chosen after seeing related outcomes, and the staircase and crossing analyses cover two slices and one curve. No result addresses generalization, training speed or modern training practice.

## 8 Related work

**Edge of stability.** Cohen et al. [6] describe the edge of stability; Damian et al. [9] explain self-stabilization of sharpness at the edge, whereas we follow what happens after the period-2 oscillation itself loses stability; Agarwala et al. [1] show progressive sharpening to the edge in second-order regression models, the approach to the regime we study beyond it; Lewkowycz et al. [21] describe a catapult phase at large rates, in which the loss first grows and then settles in flatter regions, while our maps resolve the long-run dynamics of rates where training does not settle. Further analyses include Kalra et al. [18], Chen and Bruna [4], Song and Yun [29], Chen et al. [5], Cohen et al. [7] and Hofmann [13].

**Chaos and fractals in training.** Chaos in deterministic training is discussed by Kong and Tao [19] and Jiménez-González et al. [16]; fractal trainability boundaries by Sohl-Dickstein [28], Torkamandi [31], Liu [23] and Liang and Montúfar [22]; riddled basins and reproducibility by Ly and Gong [24]. Learning dynamics in games show chaos [27] and phase locking [25], for adaptive rules rather than descent on one loss.

**Dynamical systems.** Shrimp-shaped windows are described by Gallas [10, 11]; circle maps, Arnold tongues, devil's staircases and the breakdown of invariant circles by Arnold [2], Aronson et al. [3], Curry and Yorke [8], Jensen et al. [14, 15], Ostlund et al. [26]; numerical bifurcation analysis of maps by Kuznetsov and Meijer [20]. A literature check using generic queries found no report of quasi-periodic training, Arnold tongues or rotation-number staircases for optimizers acting on one loss.

## 9 Reproducibility

Code, plans and compact receipts live in the project repository; bulky raw maps live in a separate data store referenced by SHA-256. Every figure is generated by a script in `paper/gd-route/figures/` that reads saved evidence and writes a receipt with input hashes, the generator's hash, output hashes, alt text and the values it displays. Every number in the text, and the design table, is generated by `paper/gd-route/scripts/build_numbers.py`; `data/numbers.json` records each number's source file, hash and computation, and `provenance.json` indexes all receipts. `paper/gd-route/scripts/verify.py` regenerates every figure and number and fails on any difference. The verification status of every reference is kept in `data/references-ledger.json`. The map study corresponds to the repository's EXP-028 plan and audit; its plan, receipts and audit are linked from `docs/EXP-028-results.md`.

**Placeholder.** Author contributions, acknowledgments, license choice and public code and data links are human decisions to be made before release.

## References


[1] Atish Agarwala, Fabian Pedregosa, and Jeffrey Pennington. Second-order regression models exhibit progressive sharpening to the edge of stability. In *International Conference on Machine Learning*, 2023.

[2] Vladimir I. Arnold. Small denominators. I. Mappings of the circumference onto itself. *American Mathematical Society Translations, Series 2*, 46:213–284, 1965.

[3] D. G. Aronson, M. A. Chory, G. R. Hall, and R. P. McGehee. Bifurcations from an invariant circle for two-parameter families of maps of the plane: a computer-assisted study. *Communications in Mathematical Physics*, 83:303–354, 1982. doi: 10.1007/BF01213607.

[4] Lei Chen and Joan Bruna. Beyond the edge of stability via two-step gradient updates. In *International Conference on Machine Learning*, 2023.

[5] Xuxing Chen, Krishnakumar Balasubramanian, Promit Ghosal, and Bhavya Agrawalla. From stability to chaos: Analyzing gradient descent dynamics in quadratic regression, 2023.

[6] Jeremy M. Cohen, Simran Kaur, Yuanzhi Li, J. Zico Kolter, and Ameet Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. In *International Conference on Learning Representations*, 2021.

[7] Jeremy M. Cohen, Alex Damian, Ameet Talwalkar, J. Zico Kolter, and Jason D. Lee. Understanding optimization in deep learning with central flows. In *International Conference on Learning Representations*, 2025.

[8] James H. Curry and James A. Yorke. A transition from Hopf bifurcation to chaos: computer experiments with maps on $R^2$. In *The Structure of Attractors in Dynamical Systems*, volume 668 of *Lecture Notes in Mathematics*, pages 48–66. Springer, 1978.

[9] Alex Damian, Eshaan Nichani, and Jason D. Lee. Self-stabilization: The implicit bias of gradient descent at the edge of stability. In *International Conference on Learning Representations*, 2023.

[10] Jason A. C. Gallas. Structure of the parameter space of the Hénon map. *Physical Review Letters*, 70:2714–2717, 1993. doi: 10.1103/PhysRevLett.70.2714.

[11] Jason A. C. Gallas. Dissecting shrimps: results for some one-dimensional physical models. *Physica A*, 202:196–223, 1994. doi: 10.1016/0378-4371(94)90174-0.

[12] Baptiste Goujaud, Adrien Taylor, and Aymeric Dieuleveut. Provable non-accelerations of the heavy-ball method. *Mathematical Programming*, 2025. doi: 10.1007/s10107-025-02269-2.

[13] Thomas Hofmann. The map behind the flow: Finite-step gradient descent as a dynamical system, 2026. Version 1.

[14] M. Høgh Jensen, Per Bak, and Tomas Bohr. Complete devil's staircase, fractal dimension, and universality of mode-locking structure in the circle map. *Physical Review Letters*, 50:1637–1639, 1983.

[15] M. Høgh Jensen, Per Bak, and Tomas Bohr. Transition to chaos by interaction of resonances in dissipative systems. I. Circle maps. *Physical Review A*, 30:1960–1969, 1984.

[16] Pedro Jiménez-González, Miguel C. Soriano, and Lucas Lacasa. Leveraging chaotic transients in the training of artificial neural networks. *Physical Review Research*, 8:L022032, 2026. doi: 10.1103/t5p9-kv5w.

[17] Timothy D. Jones. Topological origins of a bi-parameter periodicity hub for the Rössler attractor, 2012.

[18] Dayal Singh Kalra, Tianyu He, and Maissam Barkeshli. Universal sharpness dynamics in neural network training: Fixed point analysis, edge of stability, and route to chaos. In *International Conference on Learning Representations*, 2025.

[19] Lingkai Kong and Molei Tao. Stochasticity of deterministic gradient descent: Large learning rate for multiscale objective function. In *Advances in Neural Information Processing Systems*, 2020.

[20] Yuri A. Kuznetsov and Hil G. E. Meijer. *Numerical Bifurcation Analysis of Maps: From Theory to Software*. Cambridge University Press, 2019.

[21] Aitor Lewkowycz, Yasaman Bahri, Ethan Dyer, Jascha Sohl-Dickstein, and Guy Gur-Ari. The large learning rate phase of deep learning: the catapult mechanism, 2020.

[22] Shuang Liang and Guido Montúfar. Gradient descent with large step sizes: Chaos and fractal convergence region. In *International Conference on Learning Representations*, 2026.

[23] Yizhou Liu. Complex fractal trainability boundary can arise from trivial non-convexity, 2024.

[24] Andrew Ly and Pulin Gong. Fractal basin geometry and the limits to predictability and reproducibility in deep learning, 2025.

[25] Konstantinos Metaxas and Themistoklis P. Sapsis. Risk-sensitive learning in population games under extreme events: Bifurcations and chaotic dynamics, 2026.

[26] Stellan Ostlund, David Rand, James Sethna, and Eric Siggia. Universal properties of the transition from quasi-periodicity to chaos in dissipative systems. *Physica D*, 8:303–342, 1983.

[27] Yuzuru Sato, Eizo Akiyama, and J. Doyne Farmer. Chaos in learning a simple two-person game. *Proceedings of the National Academy of Sciences*, 99(7):4748–4751, 2002. doi: 10.1073/pnas.032086299.

[28] Jascha Sohl-Dickstein. The boundary of neural network trainability is fractal, 2024. Version 1.

[29] Minhak Song and Chulhee Yun. Trajectory alignment: Understanding the edge of stability phenomenon via bifurcation theory. In *Advances in Neural Information Processing Systems*, 2023.

[30] Sergiy Sveleba, Ivan Katerynchuk, Yaroslav Shmyhelskyy, Lucjan Pelc, Volodymyr Brygilevych, and Nazar Sveleba. Multi-parameter dynamic learning maps of neural networks trained with Adam optimiser. *Electronics and Information Technologies*, 34, 2026. doi: 10.30970/eli.34.2.

[31] Bahman Torkamandi. Mapping the edge of chaos: Fractal-like boundaries in the trainability of decoder-only transformer models, 2025.

[32] Han L. J. van der Maas, Paul F. M. J. Verschure, and Peter C. M. Molenaar. A note on chaotic behavior in simple neural networks. *Neural Networks*, 3:119–122, 1990. doi: 10.1016/0893-6080(90)90050-U.